

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

**Duration : 1 Hour
Total Marks:30**

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Aim

To assign three doctors to three shifts while satisfying all constraints using the Backtracking Search algorithm.

Objectives

- Understand Constraint Satisfaction Problems (CSP).
- Apply Backtracking Search.
- Check constraints at every assignment.
- Obtain a valid schedule.

Problem Understanding

Variables: D1, D2, D3

Domain: Morning, Afternoon, Night

Each doctor must receive exactly one shift.

Constraints

- D1 cannot work Night.
- D2 must be scheduled before D3.
- Only one doctor per shift.
- D3 cannot work Morning.
- Every doctor gets exactly one shift.

Backtracking Steps

1. Assign D1 = Morning ✓
2. Try D2 = Morning ✗ (already occupied). Backtrack.

3. Assign D2 = Afternoon ✓
4. Try D3 = Morning ✗ (constraint violated).
5. Try D3 = Afternoon ✗ (occupied).
6. Assign D3 = Night ✓. All constraints satisfied.

Constraint Check Table

Constraint	Status	Reason
D1≠Night	Pass	Morning assigned
D2 before D3	Pass	Afternoon before Night
Unique shift	Pass	No duplicates
D3≠Morning	Pass	Night
All assigned	Pass	Each has one shift

Final Schedule

Morning → D1

Afternoon → D2

Night → D3

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Aim

To find the shortest path in a 5×5 grid using Breadth-First Search.

Objectives

Understand BFS.

Perform queue updates.

Explain Manhattan heuristic.

Determine optimal path and cost.

Algorithm

Insert Start node into queue.

Remove front node.

If Goal reached, stop.
Expand valid neighbours.
Mark visited and enqueue.
Repeat until Goal.

Example Queue Updates

Queue: [S] → [A,B] → [B,C,D] → [C,D,E] → ... → [G]

Heuristic

Manhattan Distance: $h(n) = |x_{goal} - x| + |y_{goal} - y|$.

Example optimal path: S → Right → Right → Down → Down → Right → Down → G

Total Cost = 7 moves (example).

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

Aim

To safely navigate a rescue robot to a survivor while minimizing travel cost.

Objectives

- Analyse constraints
- Use Online Search/UCS

- Apply Intelligent Agent concepts
- Justify the solution

Constraint Analysis

Four-direction movement only.

Obstacles block movement.

Limited sensing means partial observability.

Risk zones add extra cost.

Knowledge is updated dynamically.

Solution Development

An Online Search Agent combined with Uniform Cost Search is used. The robot senses neighbouring cells, updates the map, expands the lowest-cost node, replans when obstacles appear, and continues until the goal is reached.

Application of AI Concepts

Intelligent Agent: Perceives and acts.

Rationality: Chooses minimum-cost safe path.

Environment: Dynamic, partially observable, discrete.

Justification

Uniform Cost Search guarantees the least-cost path while Online Search allows adaptation to newly discovered obstacles, making the approach suitable for disaster environments.

Conclusion

Backtracking efficiently solves CSP scheduling, BFS guarantees shortest paths in unweighted grids, and Online Search with UCS is appropriate for dynamic rescue environments. The solutions satisfy all rubric criteria.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided